FENG-AN (Alan) HSIEH

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Summary

Pre-silicon modeling engineer with 4 years of experience building C++/SystemC and TLM-2.0 models and virtual platforms for high-throughput SoCs. Own 6 IP models end to end, aligned to RTL specifications and verified against RTL through HW/SW co-simulation, delivered to firmware teams and external customers. Strong in Python automation and in building AI-agent workflows that shorten model delivery and regression cycles.

Experience

SiliconMotion – ESL Modeling Team **Zhubei, Hsinchu**
Chief Engineer *July 2026 – Present*

- Independently own 6 C++/SystemC TLM IP models, written from RTL design specifications and delivered to 5+ firmware engineers; the models are also packaged into complete virtual platforms for external customers.
- Architected the data-path and control-path models of a high-throughput controller: command arbitration, DMA, error handling, command decode, and a shared inter-engine messaging IP that handshakes with every engine.
- Built an AI agent harness and reusable agent skills (Claude Code, Cursor) for spec-to-model scaffolding, documentation sync, and automated regression, cutting new model delivery from ∼2 weeks to 2–3 days.
- Mentor engineers on TLM-2.0 modeling standards and development workflow.

Senior Engineer *November 2022 – June 2026*

- Developed and maintained C++/SystemC TLM models of peripheral and controller IPs against RTL design specifications; ran HW/SW co-simulation with DV patterns and verified models against RTL, resolving CModel and firmware bugs at the HW/FW boundary.
- Built and maintained full-system virtual platforms on Synopsys VDK, integrating processor fast models with multiple TLM models, so firmware teams could bring up software months ahead of silicon.
- Profiled and tuned processor fast models and platform integration to improve simulation speed and stability.
- Automated RTL waveform (FSDB/VCD) parsing with Python and shell for model verification, speeding up test creation by 3x and saving ∼1 week of manual effort per cycle.
- Built Python tools for platform trace analysis and debugging, shortening issue isolation for firmware teams.

Skills

Languages & Modeling: C/C++, SystemC, TLM-2.0, Python, Shell, Verilog, Assembly
Pre-silicon & Verification: Virtual Prototyping (Synopsys VDK), HW/SW Co-simulation, RTL-vs-model Verification, Waveform Analysis (FSDB/VCD), Git
Protocols & Architecture: AMBA (AXI, AHB, APB, CHI), DMA, Arbitration, Inter-engine Messaging
Automation & AI: Claude Code, Cursor, Agent Skills, Automated Test Generation, Regression Flows
Languages: Mandarin (Native), English (TOEIC 780)

Education

National Chung Cheng University **Chiayi, Taiwan**
M.E. in Electrical Engineering *August 2020 – July 2022*
Thesis: An End-To-End System For Road Agent Behavior Classification (Deep Learning, Computer Vision)

National University of Kaohsiung **Kaohsiung, Taiwan**
B.E. in Computer Science and Information Engineering *September 2016 – June 2020*